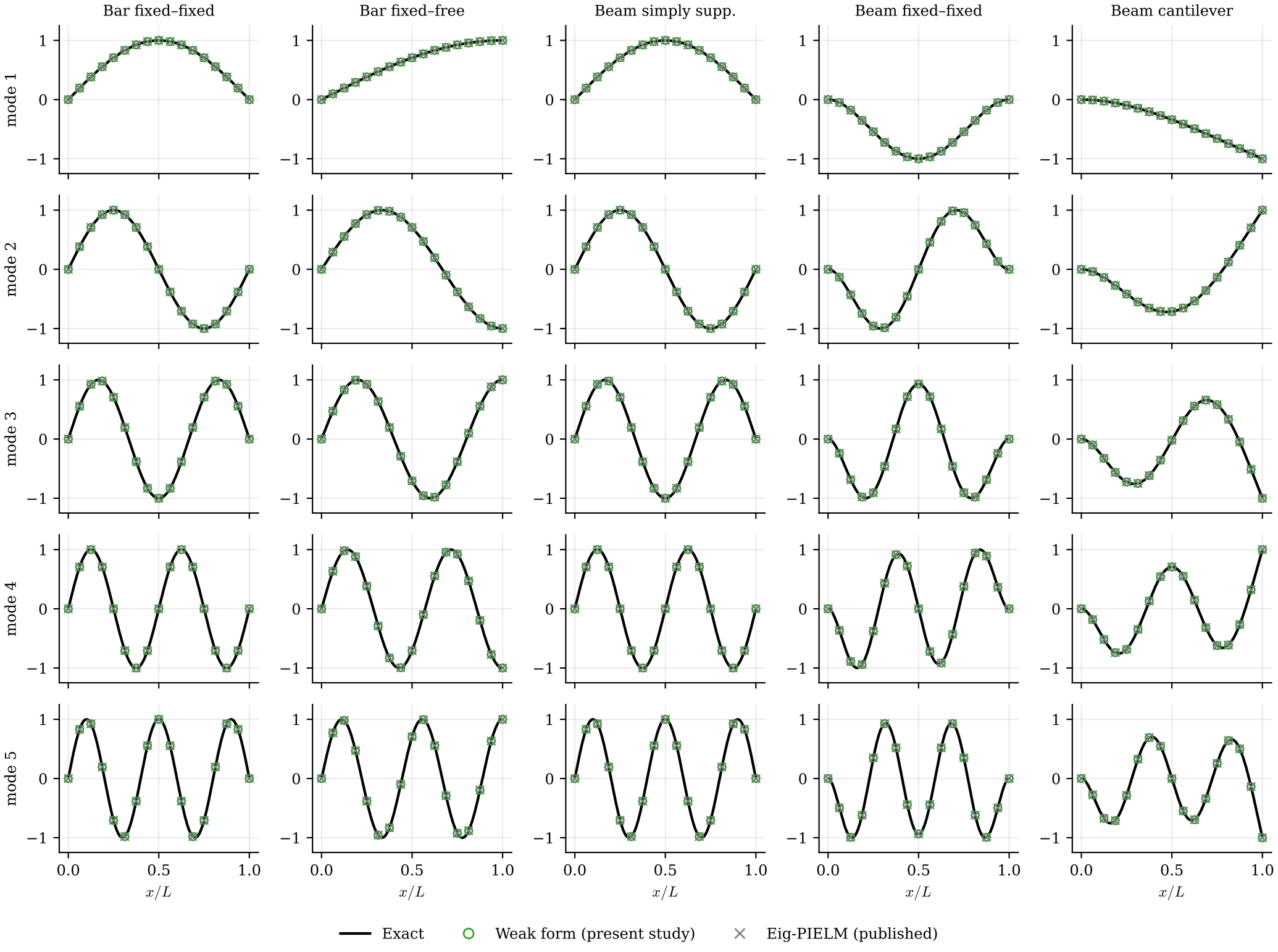

Bar fixed–fixed
Bar fixed–free
Beam simply supp.
Beam fixed–fixed
Beam cantilever
mode 1
mode 2
mode 3
mode 4
mode 5
1
0
−1
0.0
0.5
1.0
x/L
Exact
Weak form (present study)
Eig-PIELM (published)